Watershed Segm Algorithm → Rail fill valley

2,3,4,[9,1,0]

↳ used in object identification when multiple single objects

example

2	5	6	2	1
1	2	3	3	3
6	0	2	0	5
5	1	4	8	7
3	7	3	9	8

→ Both A and B are valley

1) Add 1 to 0 and classify as A

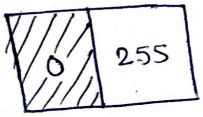
2) on boundary ⇒ watershed

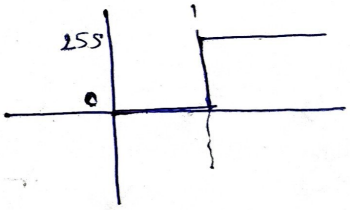
Matching ⇒ Interest point matching

↓

where most features match

detect edge





- Gaussian → smooth, remove noise, find interest point

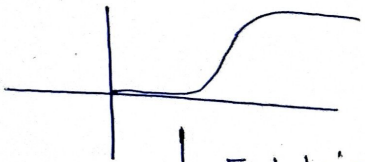
↓ gaussian

→ can be edges, corners, blob

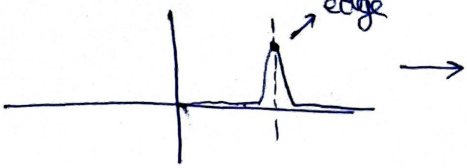
Interest point

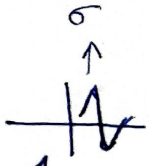
↳ changes in intensity

↳ have information

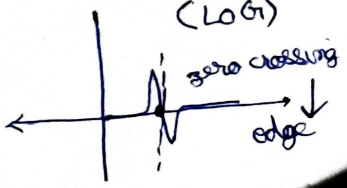


↓ First derivative

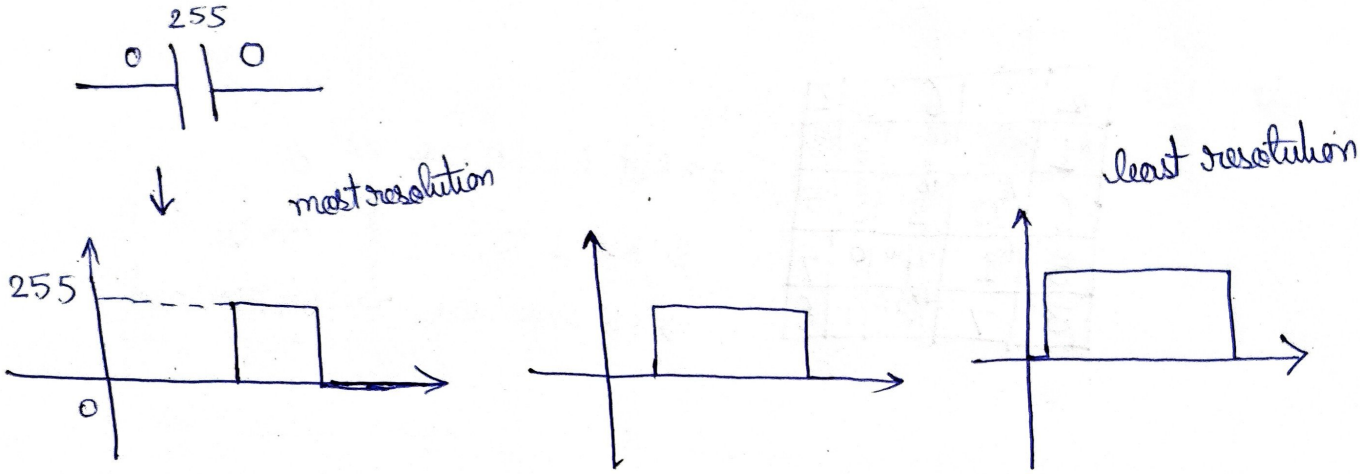




second derivative (LOG)

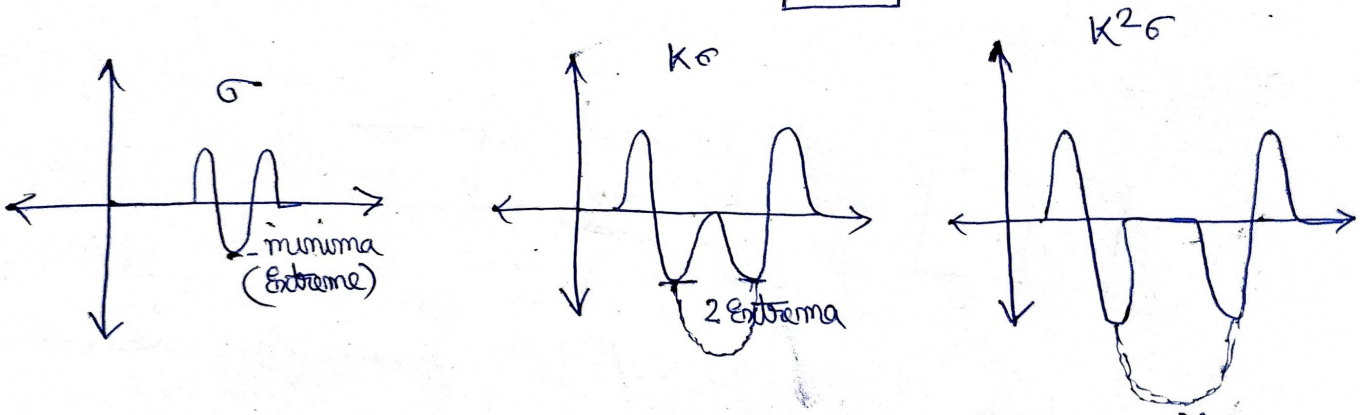


2) Blob ⇒ 2 edge or more
(4)



aspect ratio
↓
change resolution
↓
Second Derivative (LOG)

$K > 1$



we decrease the value of sigma σ so all work on different resolution

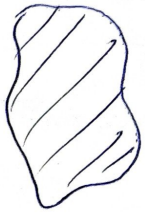
SIFT

Scale
Invariant
Feature
Transform

LOG ≈ DOG → Difference of gaussian

$G_{K\sigma}(x,y) - G_{\sigma}(x,y)$

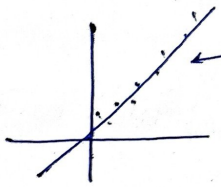
here we compare with it's neighbour with above and below points too compare with 26x26 points



Boundary → Edgedetection ⇒ lines, curves, Circular, etc

→ Hough Transform → lines

Feature Extraction / Boundary detection Techniques



best fit line from points

lines

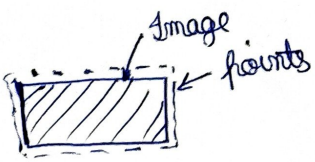
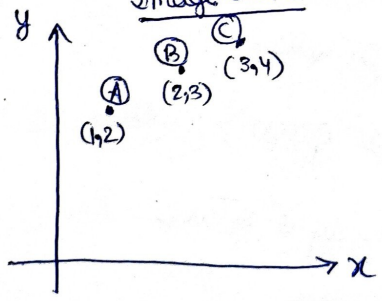
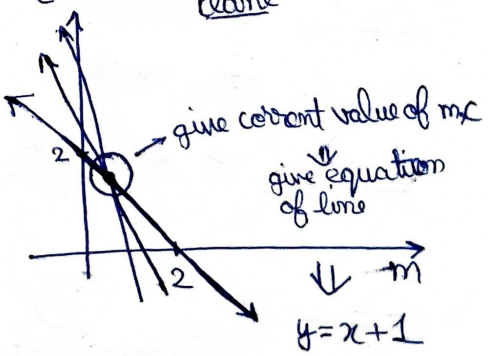


Image Plane



Parameter Plane



give covarent value of mx

give equation of line

$y = mx + c$ $A = (1,2)$

$y = mx + c$

$2 = m + c$

$y = x + 1$

$$s = \frac{1}{\frac{x_1 - x_2}{y_1 - y_2}}$$

$s = 1$

$y = mx + c$

$2 = 1 \times 1 + c$

$c = 1$

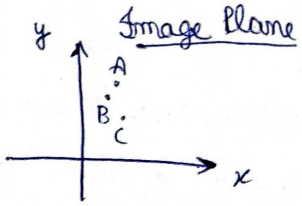
$y = 1x + 1$

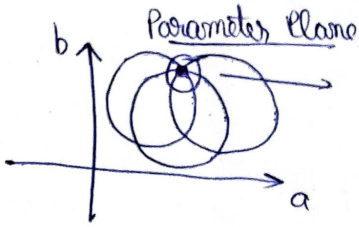
*A point in image plane ⇒ line in parameter plane

↳ Circles → fix radius

$$(x_i - a)^2 + (y_i - b)^2 = r^2$$

↳ constant





↓ we got the equation of circle

Generalised hough Transform :- fix reference point

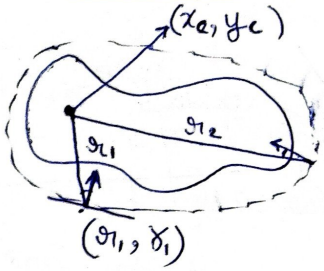


image → parameter

(x_c, y_c) (r_1, γ_1)

↓ distance equation

SIFT :-

step-1 → scale space Extrema detection

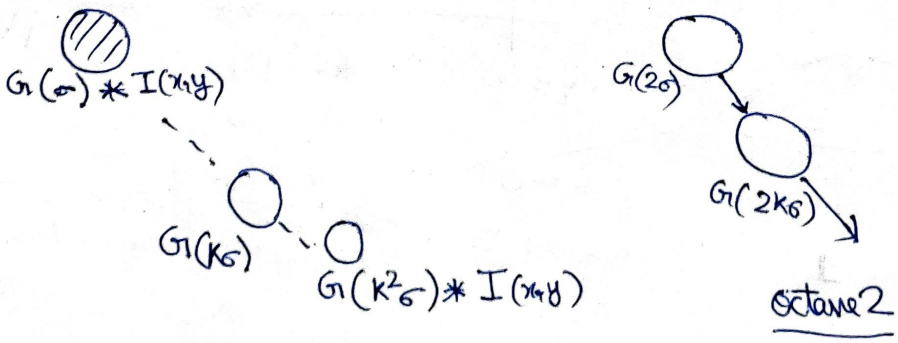
step-2 → determine location and scale at each candidate location

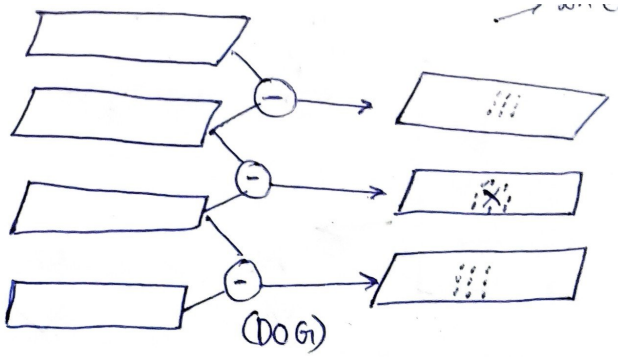
step-3 → orientation estimation

step-4 → Keyfloard description

multiple octaves

step1 ⇒





(DoG)

when x is extrema

$$DoG_1 = \overset{1st}{G(\sigma) * I(x,y) - G(k\sigma) * I(x,y)}$$

$$\overset{2nd}{=} G(k\sigma) * I(x,y) - G(k^2\sigma) * I(x,y)$$

⇒ Point X is point of interest only if it is extrema

step 2

(x, y, σ)

↳ Taylor series expansion to determine the extrema

↓

$\hat{S} = (x, y, \sigma)$ ⟶ $D(x, y, \sigma)$ ⇒ $|D(x,y,\sigma)| < 0.03$

↳ potential location ↳ then discarded for being of low contrast.

2nd order

$$f(x+h) = f(x_0) + h f'(x) + h^2 \frac{f''(x)}{2} + \dots$$

step 1 and step 2 ⇒ scale invariant

step 3 ⇒ we get gradient and angle ⇒ to make orientation independent

↓

HOG → Histogram of (calculated) gradient

Pixel at location (2,2) = 20

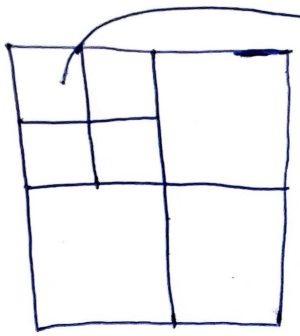
$$G_x = I(x+1, y) - I(x-1, y)$$

$$G_y = I(x, y+1) - I(x, y-1)$$

$$M_G = \sqrt{G_x^2 + G_y^2}$$

$$\theta = \tan^{-1} \frac{G_y}{G_x}$$

SIFT descriptor



16×16

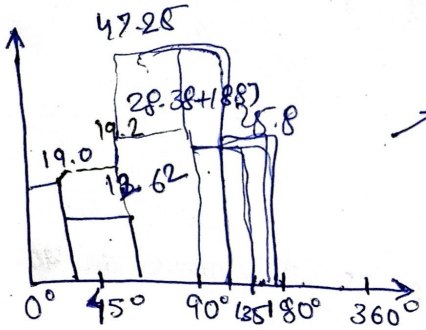
$$\begin{bmatrix} 10 & 30 & 10 & 10 \\ 10 & 20 & 40 & 30 \\ 70 & 40 & 50 & 10 \\ 10 & 60 & 10 & 10 \end{bmatrix}$$

→ M ⇓ , θ ⇓

$$\begin{bmatrix} - & - & - & - \\ - & 31.6 & 41.2 & - \\ - & 44.7 & 42.4 & - \\ - & - & - & - \end{bmatrix} \quad \begin{bmatrix} - & - & - & - \\ - & 18° & 76 & - \\ - & 116° & 225° & - \\ - & - & - & - \end{bmatrix}$$

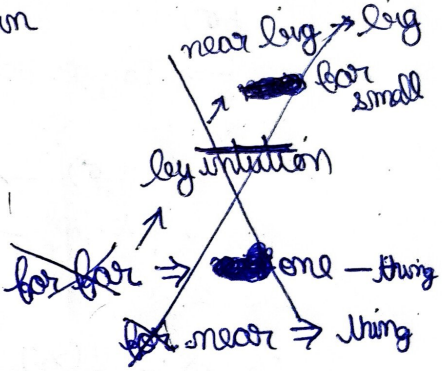
only in 0° to 360° plane

make histogram ↙



→ 8×4×4 = 128

→ 8 bin



~~for 45°~~

$$\text{~~for 45°~~} \left(1 - \frac{45-18}{45}\right) \times 31.6 \Rightarrow 40\% \times 31.6$$

$$\text{for } 45° \Rightarrow \left(1 - \frac{45-18}{45}\right) \times 31.6 \Rightarrow \frac{2}{5} \times 31.6 \Rightarrow$$

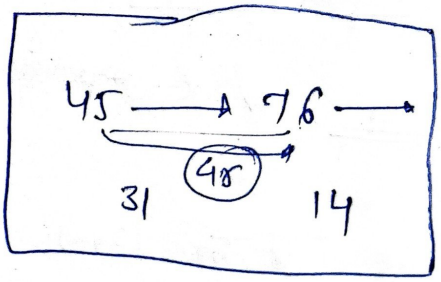
use unitary law

$$\text{for } 90° \Rightarrow \left[1 - \frac{(90-\cancel{5})}{90}\right] \times 41.2 \Rightarrow 34.6$$

$$\frac{90-76}{90} \times 41.2 \Rightarrow 6.4$$

$$\Rightarrow 28.38$$

$$1 - \frac{90-76}{90} \quad \Rightarrow 12.81$$



$$\frac{31}{48} \qquad \frac{14}{48}$$

$$\frac{76}{90} \qquad \frac{14}{90} \times \qquad \frac{14}{48} \qquad \frac{31}{48} \times 41.2$$